Air University

AIR UNIVERSITY

Digital Logic Design (Lab)

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Experiment No. 02

Title: Verification of DeMorgan’s Law

Objectives:

- To investigate and confirm DeMorgan's Theorem using truth tables.
- To reveal the intriguing equivalence between NAND/NOR gates and their AND/OR counterparts through logic functions.
- To understand the importance of DeMorgan's Laws in creating efficient digital circuits.

Equipment and Components:

- Breadboard
- Logic gate ICs 7408 (AND), 7432 (OR), 7404 (NOT)
- 5V power supply
- Connecting wires
- Optional: Multimeter

Introduction:

Welcome to the fascinating field of Boolean algebra, where we engage with the captivating realm of logical operations! Central to this investigation is DeMorgan's Theorem, an insightful contribution from Augustus DeMorgan. This theorem illustrates the graceful interaction between AND and OR operations through negation. Consider this: DeMorgan’s First Law indicates that inverting the outcome of an AND operation is equivalent to applying an OR operation to the negated inputs. Conversely, DeMorgan’s Second Law transforms OR operations, enabling us to view them as AND operations involving negated inputs. These concepts extend beyond theoretical discussions; they are essential instruments for simplifying intricate digital circuits and mastering circuit design. In this experiment, we will put these ideas into action, verifying them through practical activities and analysis of truth tables.

Procedure:

1. **Set up the Breadboard:** Begin by connecting the 5V power supply to your breadboard, laying the groundwork for our experiment.
2. **Position Logic Gate ICs:** Carefully insert the AND (7408), OR (7432), and NOT (7404) ICs into your breadboard.
3. **Connect Power and Ground:** Verify that each IC is powered according to its requirement.
4. **Wire Inputs and Outputs:**
   - For the AND gate: Attach two input switches, prepared to test all conceivable combinations.
   - For the OR gate: Make similar connections with two input switches.
   - For the NOT gate: Connect a single input switch since it only requires one input.
5. **Analyze Input Combinations:**
   - Systematically evaluate each possible combination for both 2-input gates:
     - Test all four combinations (00, 01, 10, 11) for AND and OR gates.
     - Test both possibilities (0 or 1) for the NOT gate.

6. **Log Outputs:** Collect your observations by noting the outputs for each combination using a multimeter or LED indicators for added visual appeal.
7. **Construct Truth Tables:** Develop truth tables for all three operations based on your documented outputs:
    - Create tables showing inputs vs. outputs to verify how each gate behaves according to its logical operation rules.

Logic Expression and Truth Table:

DeMorgan's First Law

- **Expression:** ¬(A · B) = ¬A + ¬B
- **Truth Table:**
  LHS:

A	B	A · B	¬(A · B)
0	0	0	**1**
0	1	0	**1**
1	0	0	**1**
1	1	1	**0**

RHS:

¬A	¬B	¬(A + B)
1	1	**1**
1	0	**1**
0	1	**1**
0	0	**0**

DeMorgan's Second Law

- **Expression:** ¬(A + B) = ¬A · ¬B
- **Truth Table:**
  LHS:

A	B	A + B	¬(A + B)
0	0	0	**1**
0	1	1	**0**
1	0	1	**0**
1	1	1	**0**

RHS:

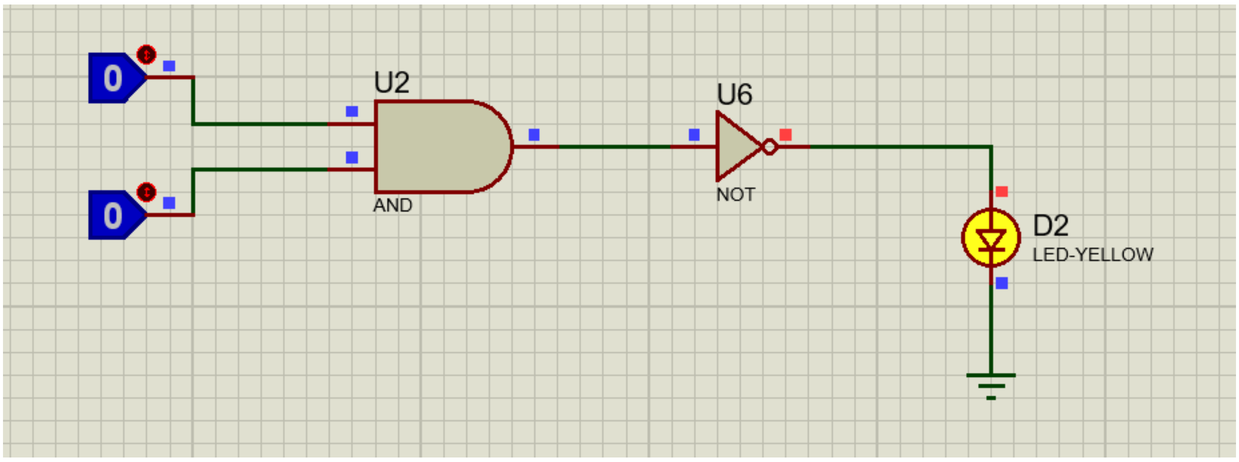
¬A	¬B	¬A · ¬B
1	1	**1**
1	0	**0**
0	1	**0**
0	0	**0**

Simulated Results:

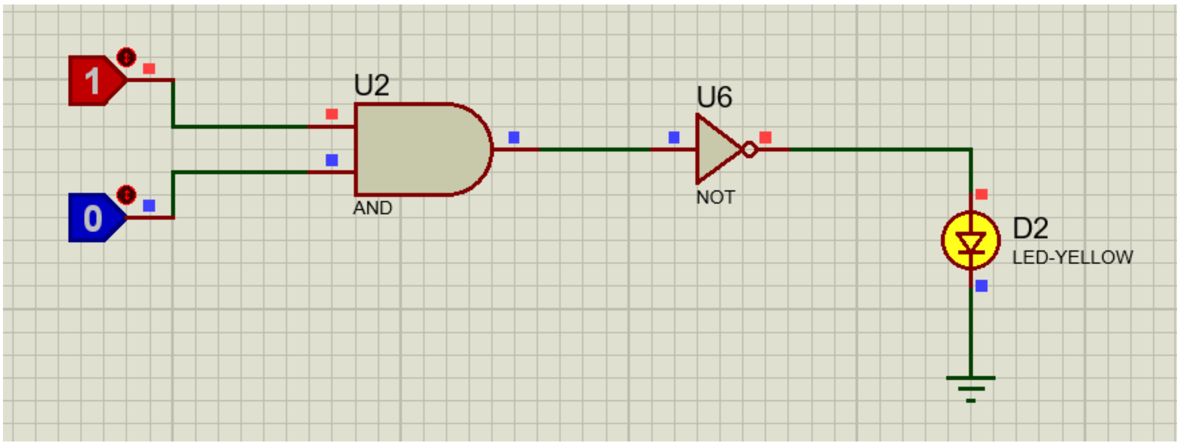
DeMorgan's First Law

LHS:

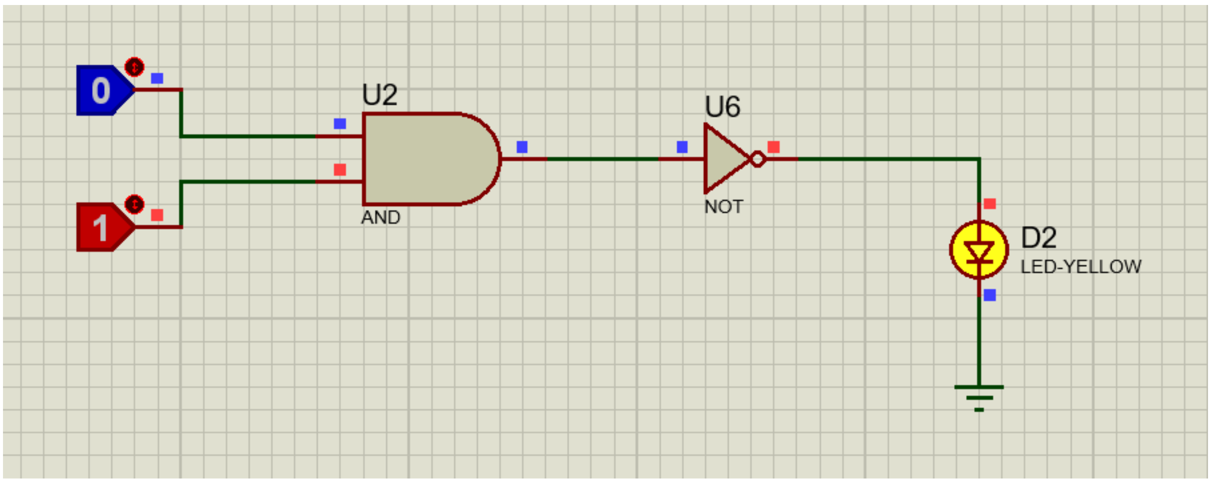
For 0 and 0 input



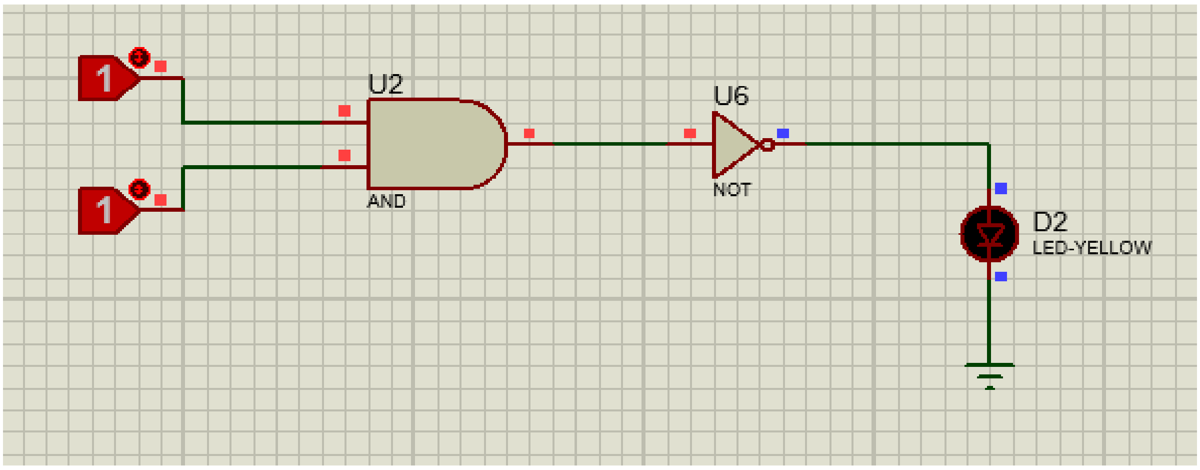
For 1 and 0 input



For 0 and 1 input

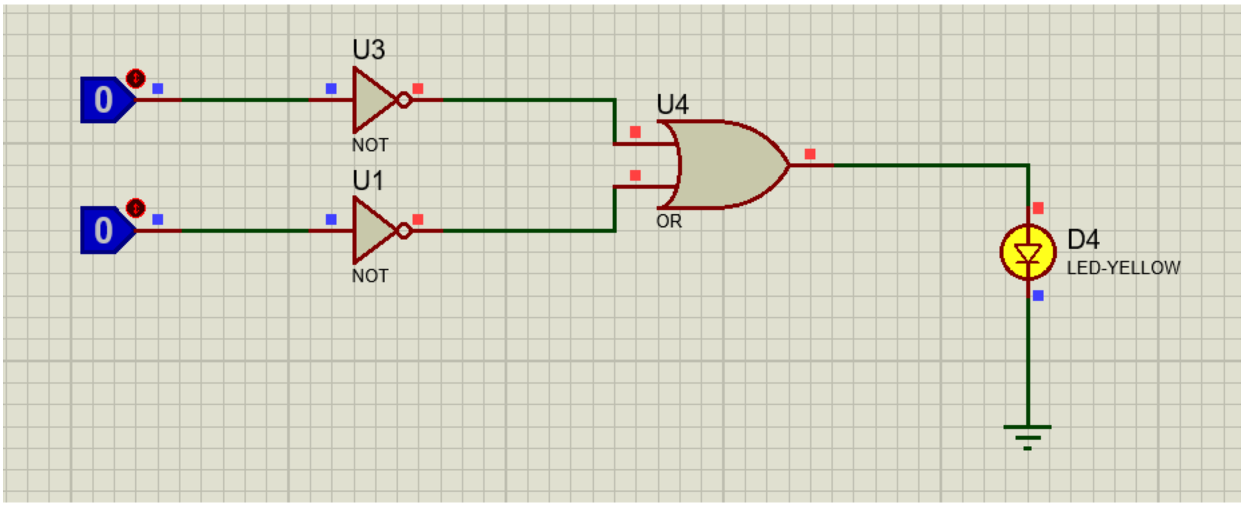


For 1 and 1 input

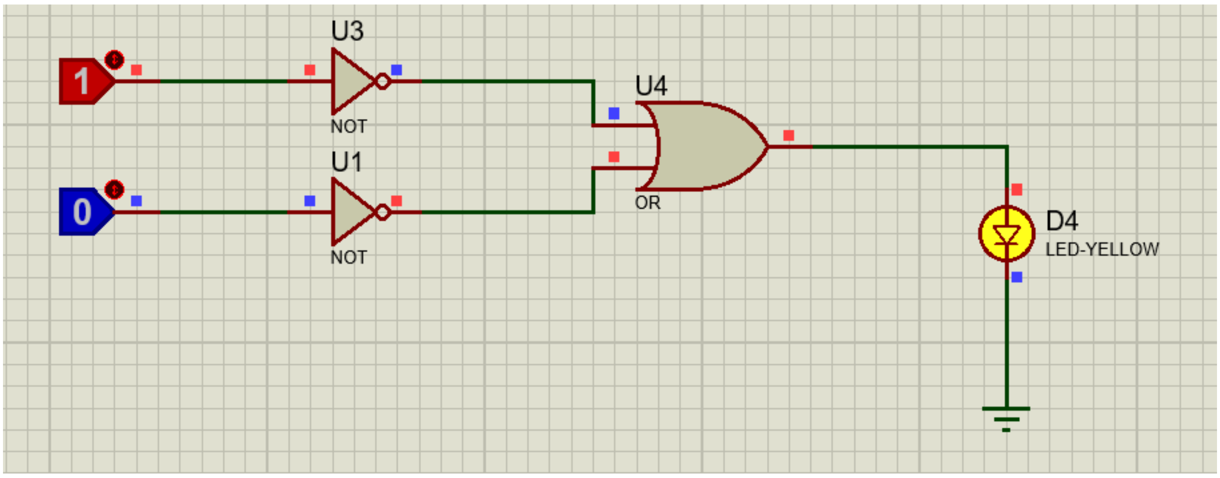


RHS:

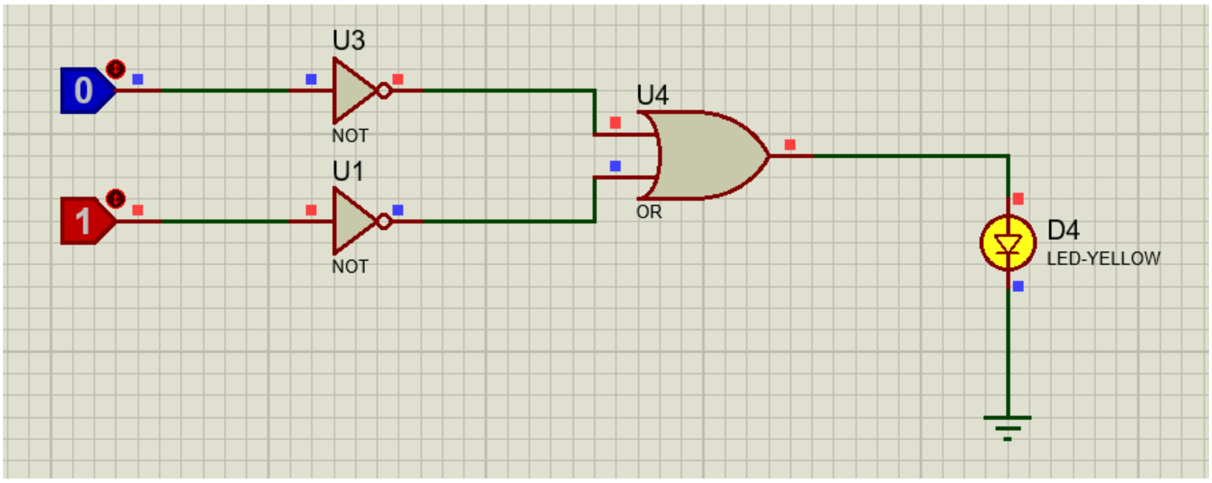
For 0 and 0 input



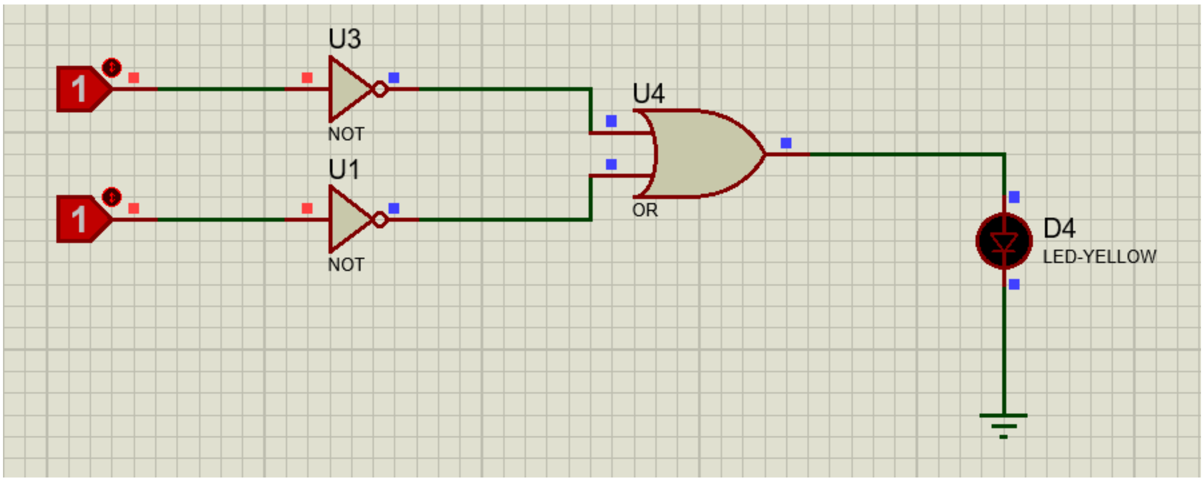
For 1 and 0 input



For 0 and 1 input



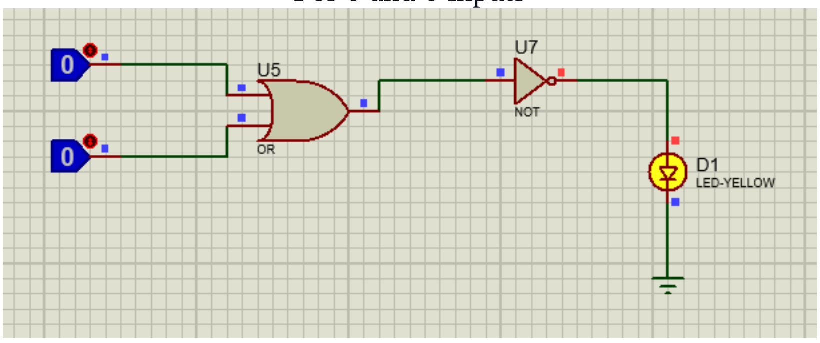
For 1 and 1 input



DeMorgan's Second Law

LHS:

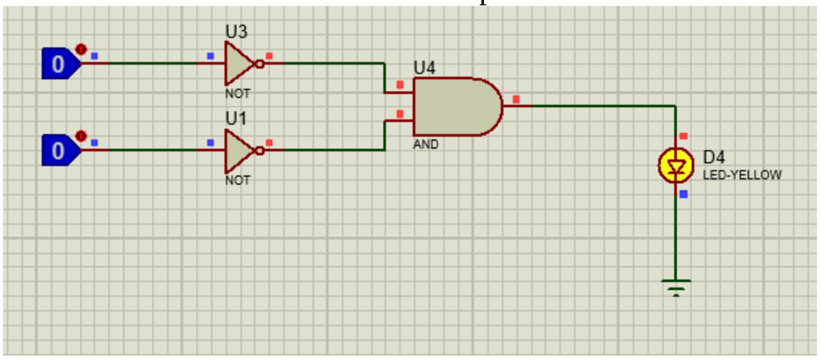
For 0 and 0 inputs



For all other inputs result will be 0 (LOW).

RHS:

For 0 and 0 input



For all other inputs results will be 0 (LOW)

Conclusions:

This experiment effectively illustrated the fundamental principles of digital logic through practical exploration and truth table examination. We experienced firsthand how AND, OR, and NOT gates function together, highlighting their significance in building complex digital circuits. A solid grasp of these basic logic gates is crucial for anyone interested in digital logic design since they act as foundational tools for circuit construction and evaluation. In the future, experiments could explore the intricacies of multi-input logic circuits or integrate additional logic functions like NAND and NOR gates to further showcase the extensive potential of these principles in the field of digital design.

THANK YOU FOR YOUR TIME!

<u>THE END</u>